

Question # 1 of 10 | Start time: 23:40:27 PM, 13 June 2019 | Total Marks: 1

The radial component of velocity for a particle moving in circular path is

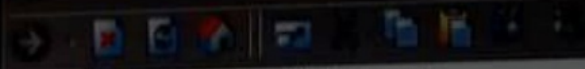
Select the correct option

- ☒ constant
- ☐ radius itself
- ☐ variable
- ☐ zero

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MTH622 Vectors and Classical Mechanics

Question No : 5 of 41

The functions in Green's theorem will be

Answer (Please select your correct option)

- ☒ Continuous derivatives
- ☐ Discrete derivatives
- ☐ Continuous partial derivatives
- ☐ Discrete partial derivatives

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MathType (Lite mode) - ...

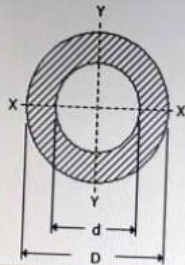
VU Examination Syste...

MTH622 Vectors and Classical Mechanics

Question No : 15 of 41

Marks: 1 (Budgeted Time 1 Min)

Moment of inertia of a hollow circular section, as shown in the below figure about X-axis, is



Answer (Please select your correct option)

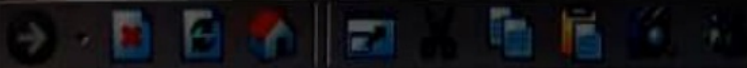
☐ $\pi/16 (D^2 - d^2)$

☐ $\pi/16 (D^3 - d^3)$

☐ $\pi/16 (D^4 - d^4)$

☒ $\pi/64 (D^4 - d^4)$

Done

Address <http://server/vutes/client/Paper.aspx>**MTH622** Vectors and Classical Mechanics

Question No : 13 of 41

Cylinder has moment of inertia given by formula of

Answer (Please select your correct option)

☐ (ml^2) ☐ mr ☐ $\frac{1}{2}(mr)$ ☐ $5(mr^2)$

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Done

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VU Examination Syste...



Question No : 19 of 41 Marks: 1 (Budgeted Time 1 Min)

The periodic time of a particle with simple harmonic motion is _____ proportional to the angular velocity

Answer (Please select your correct option)

☐ Directly

☒ Inversely

☐ Square root

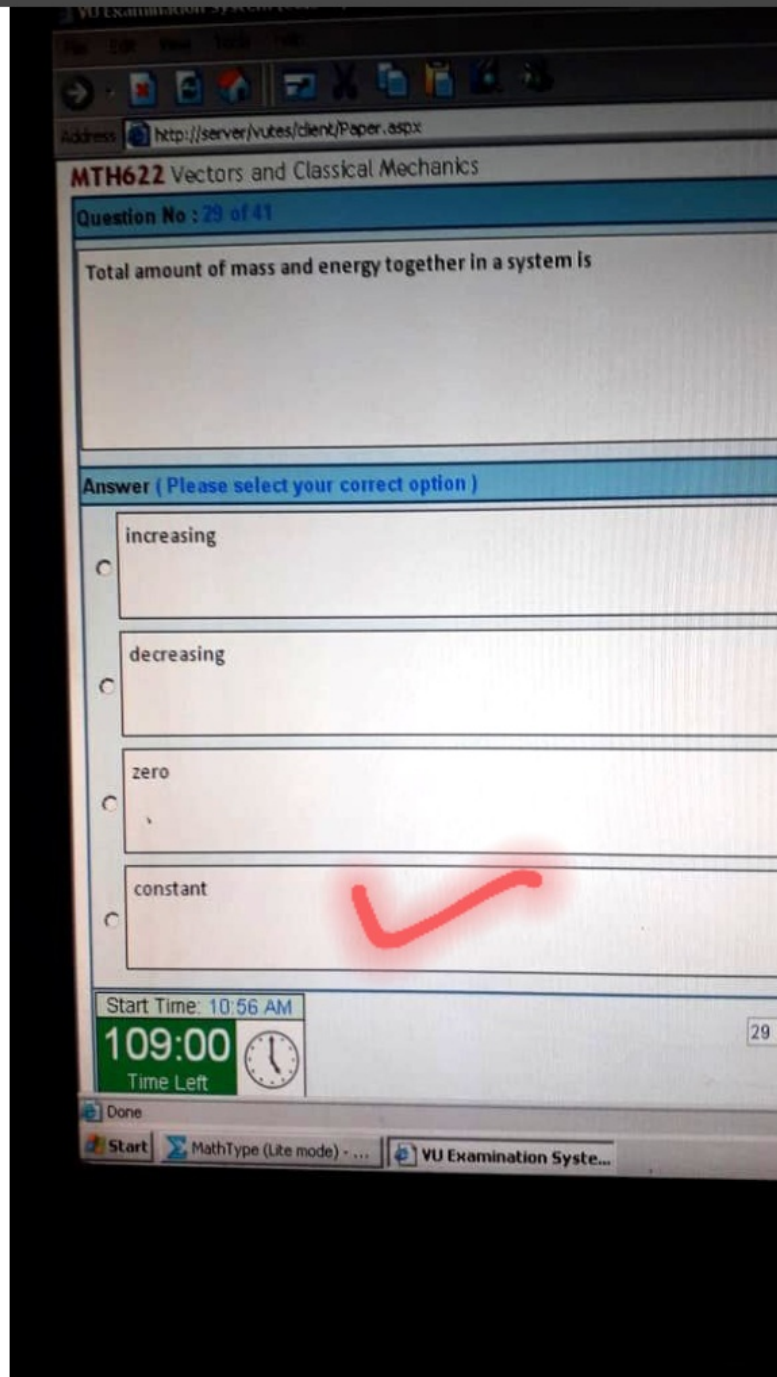
☐ None of these

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VU Examination System (VU-Client) - Vectors and Classical Mechanics

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MTH622 Vectors and Classical Mechanics

Question No : 26 of 41

The time taken by the particle to complete one oscillation is called

Answer (Please select your correct option)

- ☐ amplitude
- ☒ time period
- ☐ frequency
- ☐ none of these

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MTH622 Vectors and Classical Mechanics

Question No : 24 of 41

Marks: 1 (Bu

In a conservative field, the total energy of particle _____ throughout the motion.

Answer (Please select your correct option)

☐ varies rapidly

☐ does not remain constant

☒ remains constant

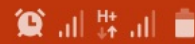
☐ none of these

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MTH622 Vectors and Classical Mechanics

Question No : 28 of 41

If there is no net force acting on body, then its acceleration is

Answer (Please select your correct option)

- ☒ zero
- ☐ constant
- ☐ increasing
- ☐ decreasing

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MTH622 Vectors and Classical Mechanics

Question No : 27 of 41

Until a force acts on a body, it's velocity is

Answer (Please select your correct option)

☒ zero

☐ constant

☐ increasing

☐ decreasing

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VU Examination Syste...

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MTH622 Vectors and Classical Mechanics

Question No : 30 of 41

Measure of average kinetic energy of molecules is

Answer (Please select your correct option)

☒ temperature

☐ energy

☐ internal energy

☐ enthalpy

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MTH622 Vectors and Classical Mechanics

Question No : 25 of 41

The path of projectile is called its

Answer (Please select your correct option)

- ☐ velocity
- ☐ acceleration
- ☒ trajectory
- ☐ none of these

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MTH622 Vectors and Classical Mechanics

Question No : 21 of 41

The rate of change of displacement of a body is called

Answer (Please select your correct option)

☒ Velocity☐ Acceleration☐ Momentum☐ None of these

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Start

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VU Examination Syste...

MTH622 Vectors and Classical Mechanics

Question No : 23 of 41

The path of a projectile in a non-resistive medium is

Answer (Please select your correct option)

☐ linear☐ Circular☒ Parabolic☐ Hyperbolic

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VU Examination Syste...

**MTH622** Vectors and Classical Mechanics

Question No : 17 of 41

At pivot point, body with mass ' m ' will have a torque by the

Answer (Please select your correct option)

☒ force☐ circle☐ central point☐ radius of circle

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Start

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VU Examination Syste...

MTH622 Vectors and Classical Mechanics

Question No : 18 of 41

Marks: 1 (Budgeted Time)

For a conservative holonomic dynamical system, the Lagrangian L , kinetic energy T and potential energy V are connected by

Answer (Please select your correct option)

☐ $L = T + V$

☒ $L = T - V$

☐ $L = 2T + V$

☐ $L = 2T - V$

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MTH622 Vectors and Classical Mechanics

Question No : 22 of 41

For an irrotational flow

Answer (Please select your correct option)

☒ $\nabla \times \vec{v} = \vec{0}$

☐ $\nabla \times \vec{v} \neq \vec{0}$

☐ $\nabla \cdot \vec{v} = \vec{0}$

☐ None of these

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Start MathType (Lite mode) - ... VU Examination System...

MTH622 Vectors and Classical Mechanics MC1

Question No : 16 of 41 Marks: 1 (Budgeted Time 1 Min)

According to parallel axis theorem, the moment of inertia of a section about an axis parallel to the axis through center of gravity (i.e. IP) is given by _____
(where, A = Area of the section, IG = Moment of inertia of the section about an axis passing through its C.G., and h = Distance between C.G. and the parallel axis.)

Answer (Please select your correct option)

☒ IP = IG + Ah²

☐ IP = IG - Ah²

☐ IP = IG / Ah²

☐ IP = Ah² / IG

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Done

Address <http://server/vutes/client/Paper.aspx>**MTH622** Vectors and Classical Mechanics

Question No : 14 of 41

Moment of inertia of sphere is given as

Answer (Please select your correct option)

☐ $2(ml^2)$

☐ r^2

☐ $1/2(r)$

☒ $2/5(mr^2)$

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VU Examination System

Address <http://server/vutes/client/Paper.aspx>

MTH622 Vectors and Classical Mechanics

Question No : 11 of 41 Marks: 1 (Budgeted Time 1 min)

The moment of inertia of a circular section of base 'b' and height 'h' about an axis passing through its vertex and parallel to base is

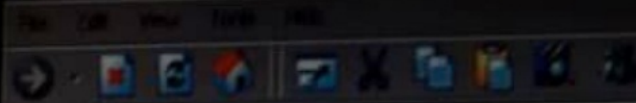
Answer (Please select your correct option)

☐ $\frac{bh^3}{12}$

☐ $\frac{bh^3}{36}$

☐ $\frac{bh^3}{3}$

☐ $\frac{bh^3}{4}$

**MTH622** Vectors and Classical Mechanics

Question No : 10 of 41

The wheels of a moving car possess

Answer (Please select your correct option)

- ☐ Potential energy only
- ☐ Kinetic energy of translation only
- ☐ Kinetic energy of rotation only
- ☒ Kinetic energy of translation and rotation both



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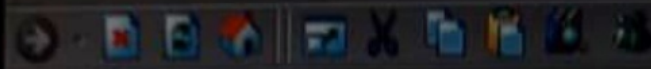


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VU Examination Syste...

**MTH622** Vectors and Classical Mechanics

Question No : 12 of 41

Moment of inertia of thin rod is given by formula of

Answer (Please select your correct option)

☒ $\frac{1}{12}(ml^2)$



☐ mr^2

☐ $2(mr)$

☐ $\frac{2}{5}(mr^2)$

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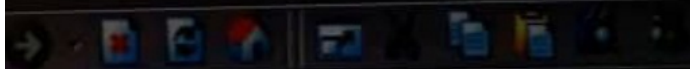
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VU Examination Syste...

Address <http://server/vutes/client/Paper.aspx>**MTH622** Vectors and Classical Mechanics

Question No : 9 of 41

The polar moment of inertia of a circular section is about

Answer (Please select your correct option)

☐ X-X axis☐ Y-Y axis☒ Z-Z axis☒ Neutral axis

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MTH622 Vectors and Classical Mechanics MC18

Question No : 5 of 41 Marks: 1 (Budgeted Time 1 Min)

If a time dependent vector function $\vec{A}$ is represented by $\vec{A}_f$ and $\vec{A}_r$ in fixed and rotating coordinate system, then

Answer (Please select your correct option)

☐ $\left(\frac{d\vec{A}}{dt}\right)_f = \left(\frac{d\vec{A}}{dt}\right)_r - \omega \times \vec{A}_r$

☒ $\left(\frac{d\vec{A}}{dt}\right)_f = \left(\frac{d\vec{A}}{dt}\right)_r + \omega \times \vec{A}_r$

☐ $\left(\frac{d\vec{A}}{dt}\right)_f = A_r - \omega \times \vec{A}_r$

☐ $\left(\frac{d\vec{A}}{dt}\right)_f = \left(\frac{d\vec{A}}{dt}\right)_r - A \times \vec{A}_r$

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MTH622 Vectors and Classical Mechanics

Question No : 7 of 41

The axes relative to which product of inertia are zero are called

Answer (Please select your correct option)

- ☒ principal axes
- ☐ principal moment of inertia
- ☐ moment of inertia
- ☐ axes of rotation

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Done

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MTH622 Vectors and Classical Mechanics

Question No : 4 of 41

The Stoke's theorem can be used to find which of the following?

Answer (Please select your correct option)

☒ Area enclosed by a function in the given region☐ Volume enclosed by a function in the given region☐ Linear distance☐ Curl of the function

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YU Examination System



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MTH622 Vectors and Classical Mechanics

Question No : 3 of 41

$$\nabla \times \nabla \varphi = ?$$

Answer (Please select your correct option)

☐ $\nabla^2 \varphi$

☐ φ

☒ 0

☐ None of these

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VU Examination Syste...

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MTH622 Vectors and Classical Mechanics MC1

Question No : 8 of 41 Mark: 1 (Budgeted Time 1 Min)

A 3×3 symmetrical matrix has ----- real eigenvalues, which may be distinct or repeated

Answer (Please select your correct option)

☐ Infinite

☐ Zero

☐ Two

☒ Three

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MTH022 Vectors and Classical Mechanics

Question No : 1 of 41

Mark

The maximum value of the directional derivative of $\varphi(x, y, z)$ is equal to the

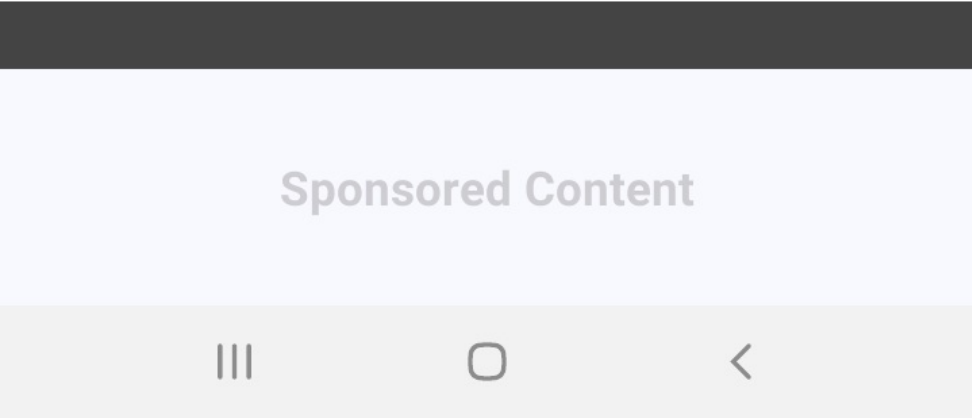
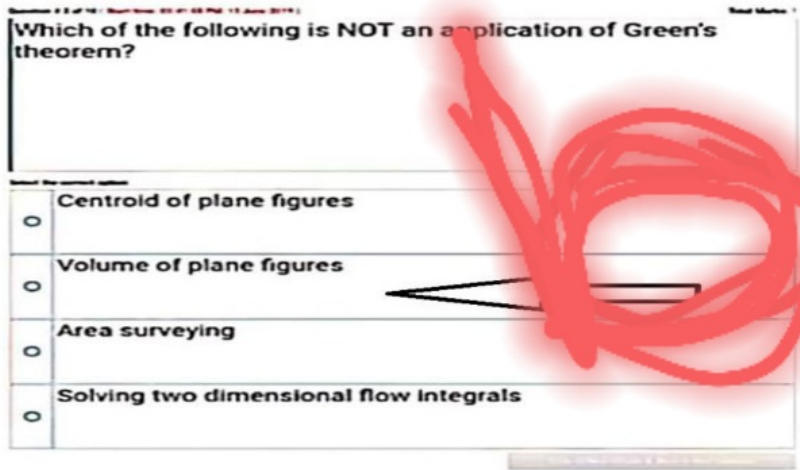
Answer (Please select your correct option)

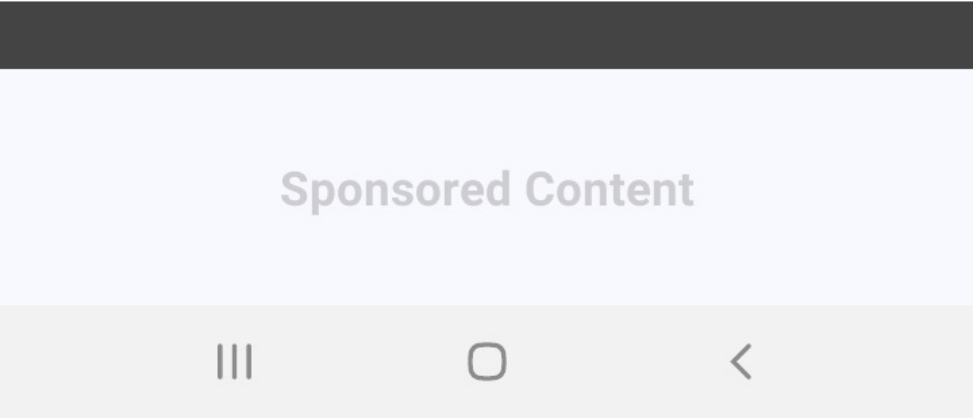
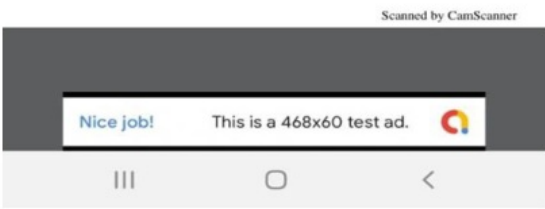
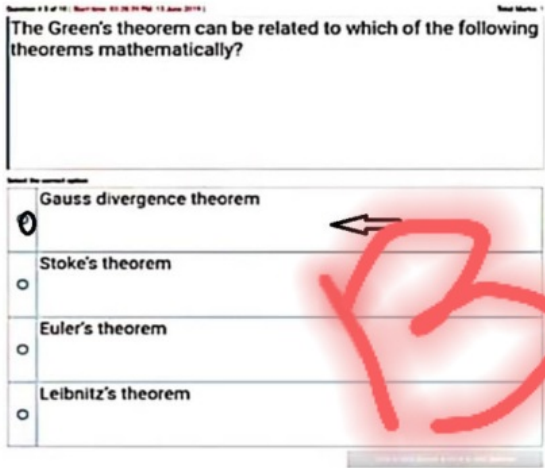
☐ Magnitude of $\nabla \times \varphi$

☐ Magnitude of φ

☒ Magnitude of $\nabla \varphi$

☐ None of these







Question 1 of 10 | Start time: 12:25 PM, 15 June 2019 | Total Marks: 1

The rate of change of tangential velocity of an object traveling in a circular orbit or path is known as

Select the correct option

☐	Normal acceleration
☐	Tangential acceleration
☐	Rectangular acceleration
☐	None of these

A hand-drawn red circle with a black arrow pointing to the 'Tangential acceleration' option.

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Nice job!

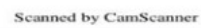
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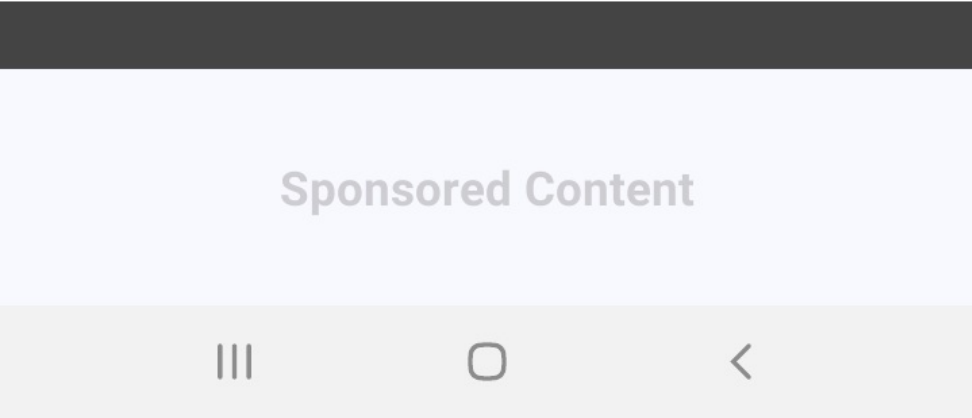
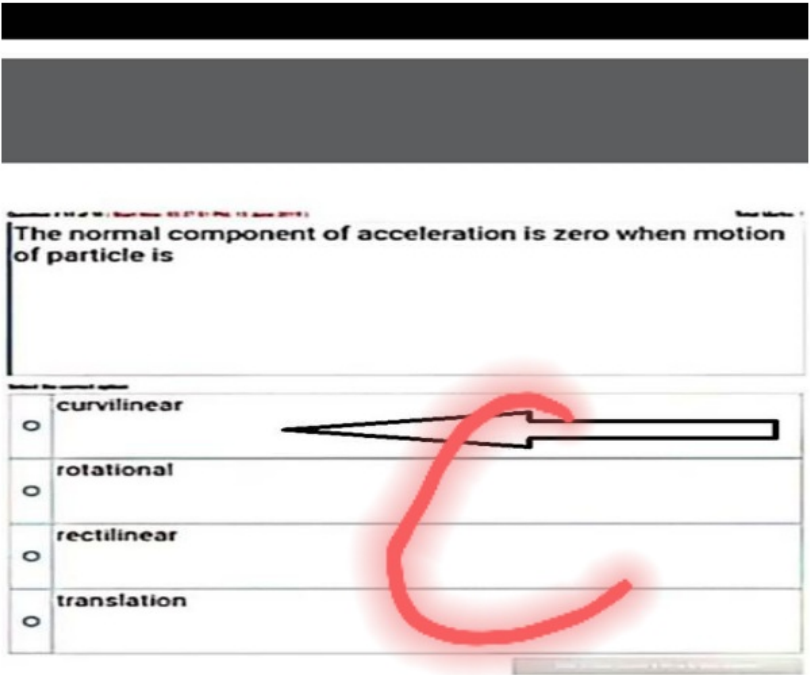


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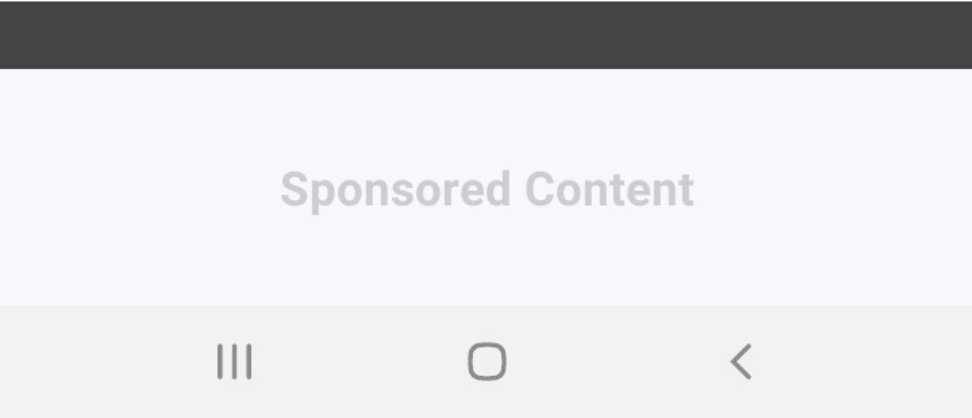
Question 1 of 10 | Question ID: 10101010 | 10 June 2019 | Your Marks: 0

Greens theorem applies only to _____ vector fields

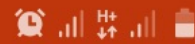
Select the correct option

☐	one dimensional
☒	two dimensional
☐	three dimensional
☐	None of these

A red handwritten letter 'b' is written over the 'two dimensional' option, with a black arrow pointing to it.



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Question 1 of 10 | Question ID: 00101 | Paid: 10 June 2019 | Total Marks: 1

The radial component of velocity for a particle moving in circular path is

Select the correct option

☐	constant
☐	radius itself
☐	variable
☒	zero

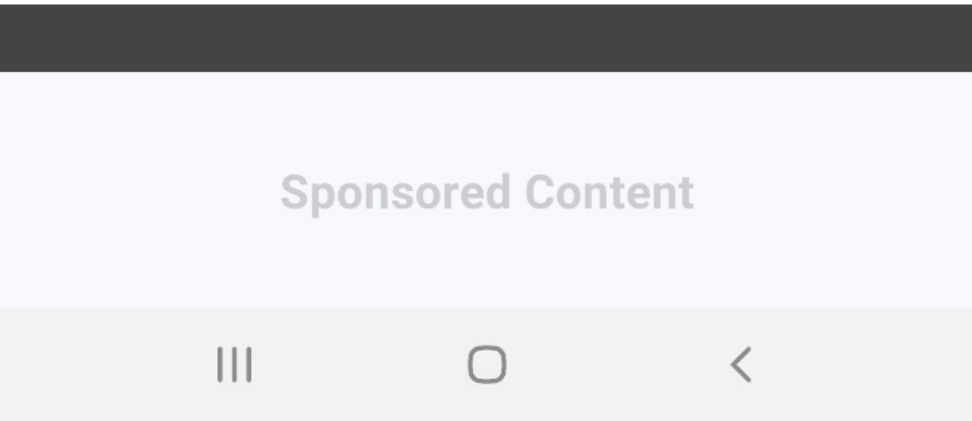
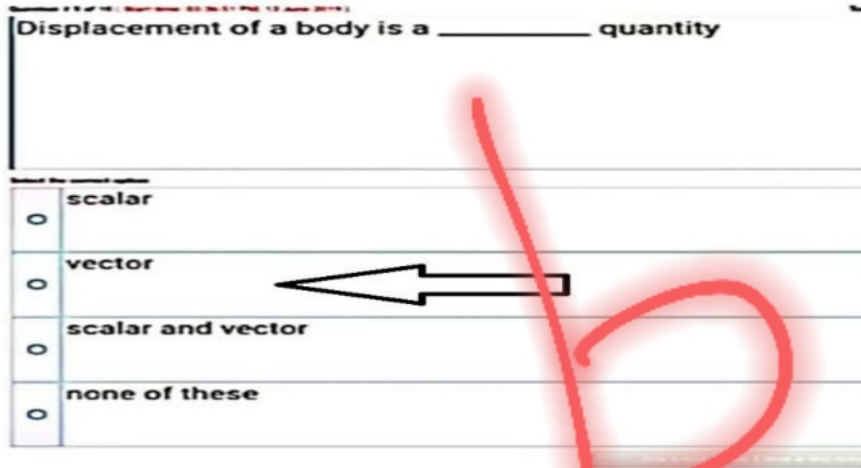


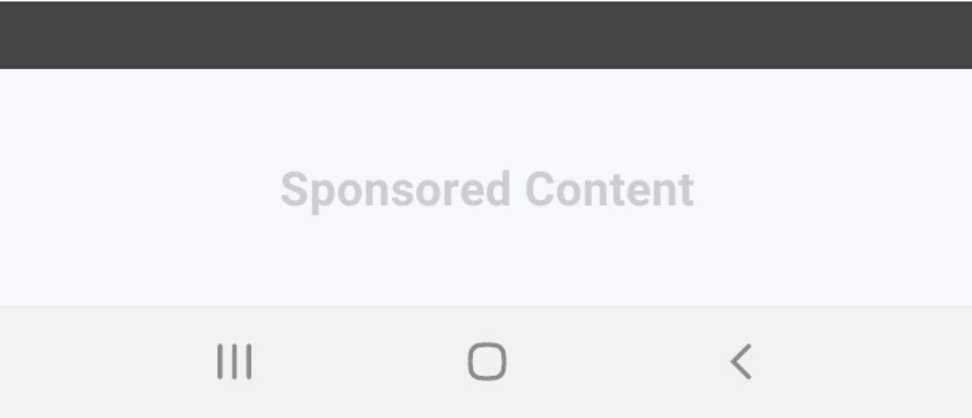
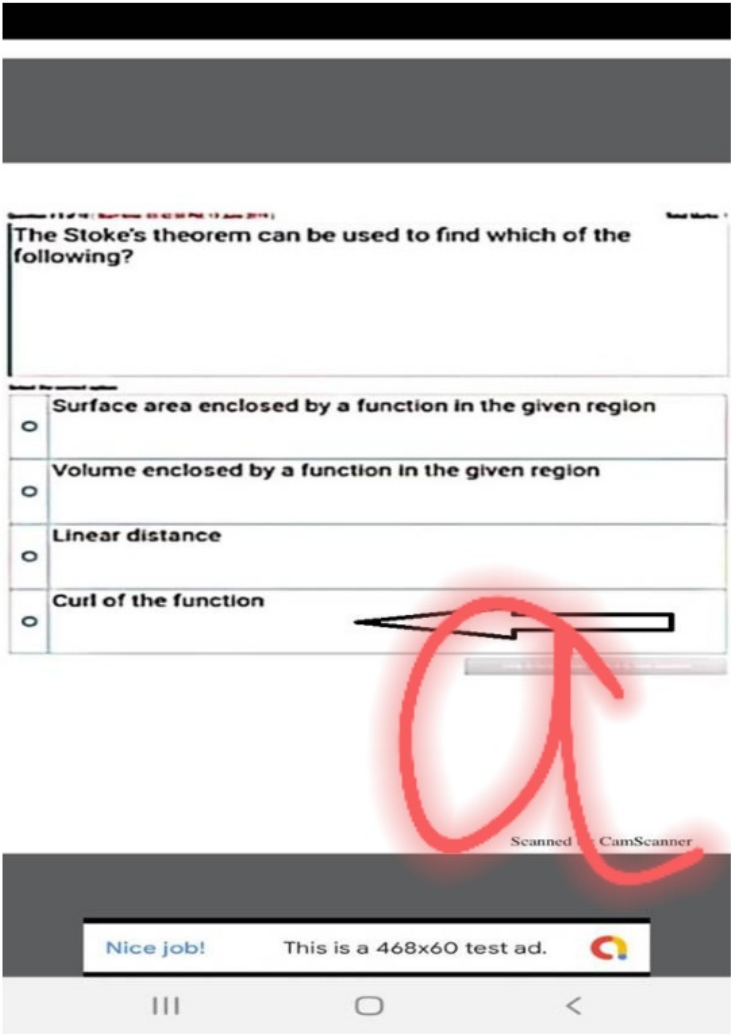
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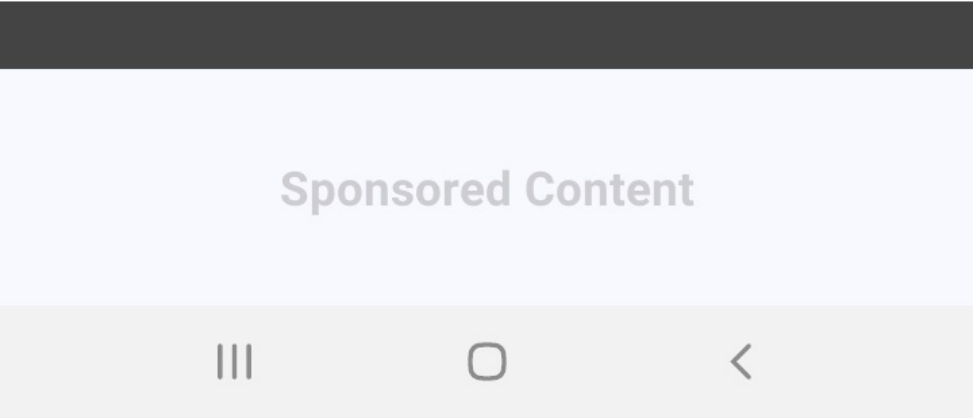
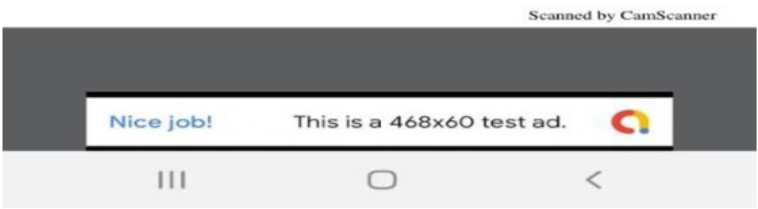
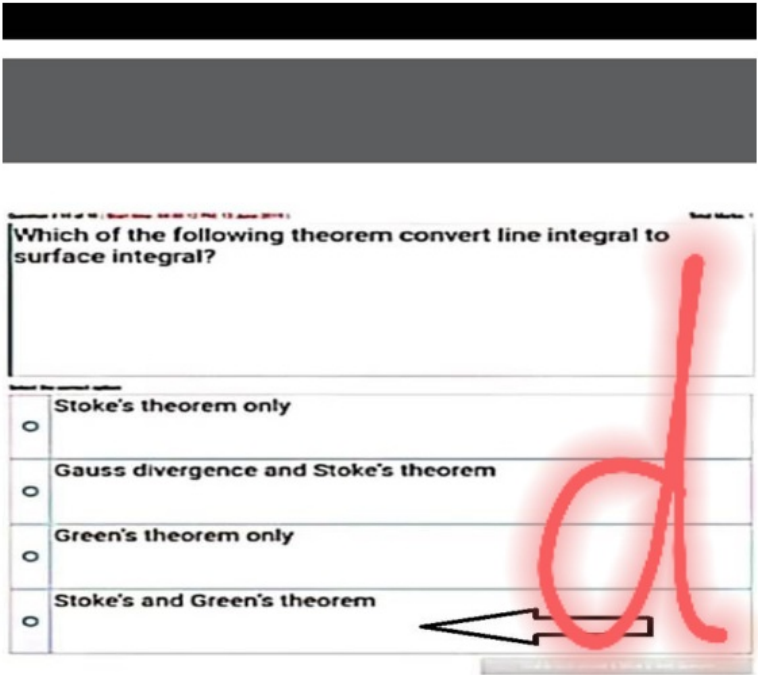


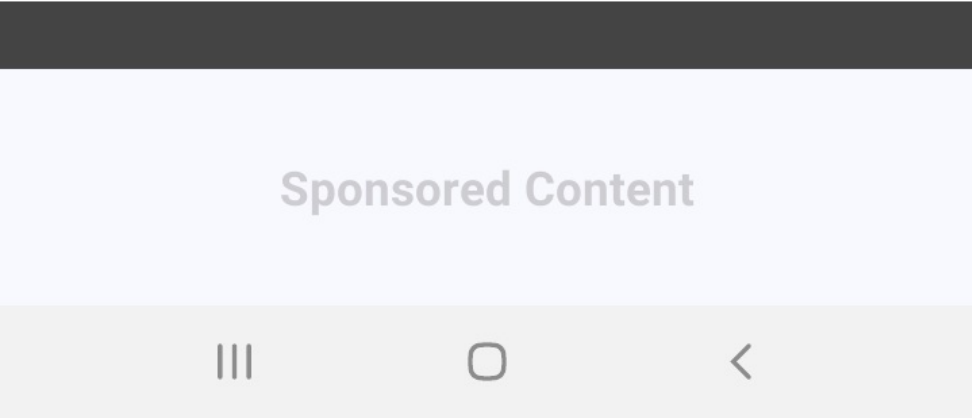
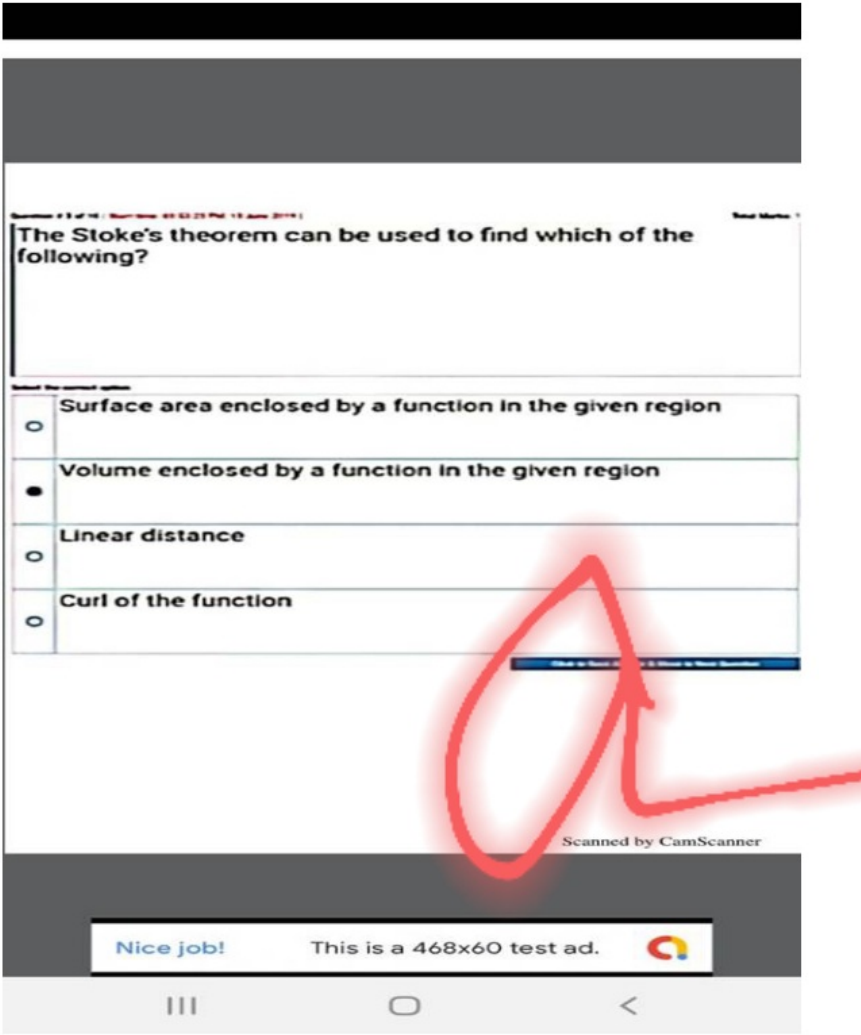
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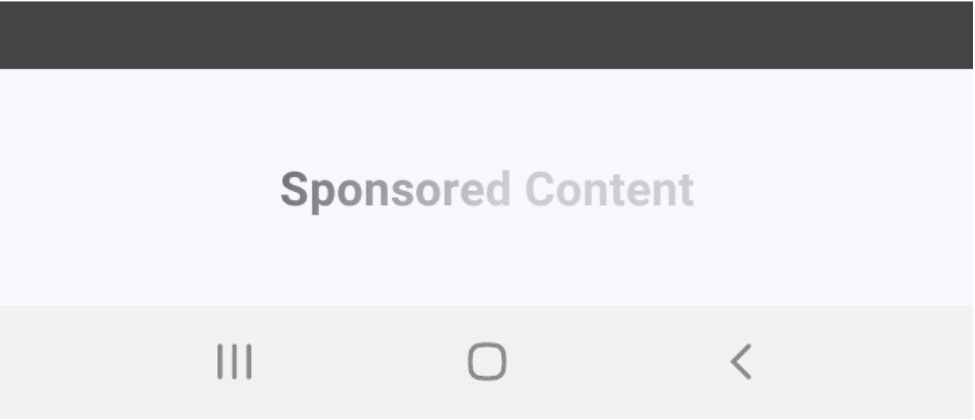
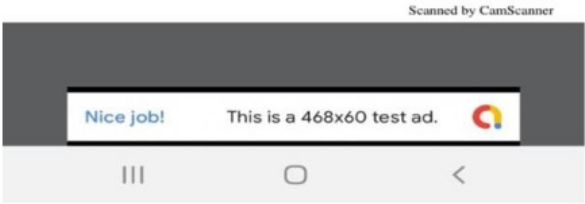
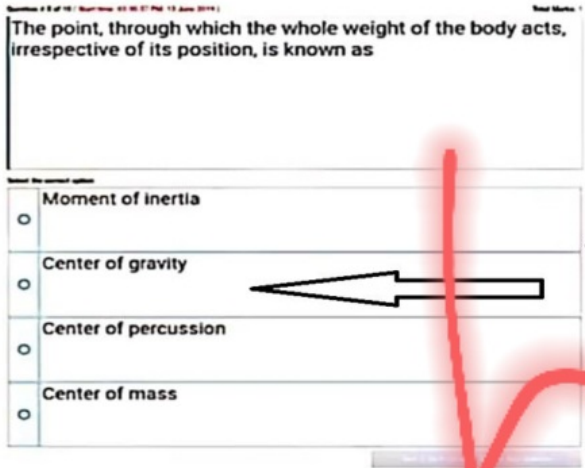




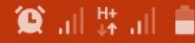








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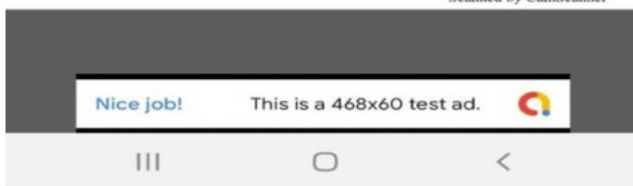
Question 4 of 10 | Start time: 43:29:25 PM, 13 June 2019 | Total Marks: 1

The rate of change of tangential velocity of an object traveling in a circular orbit or path is known as

Select the correct option

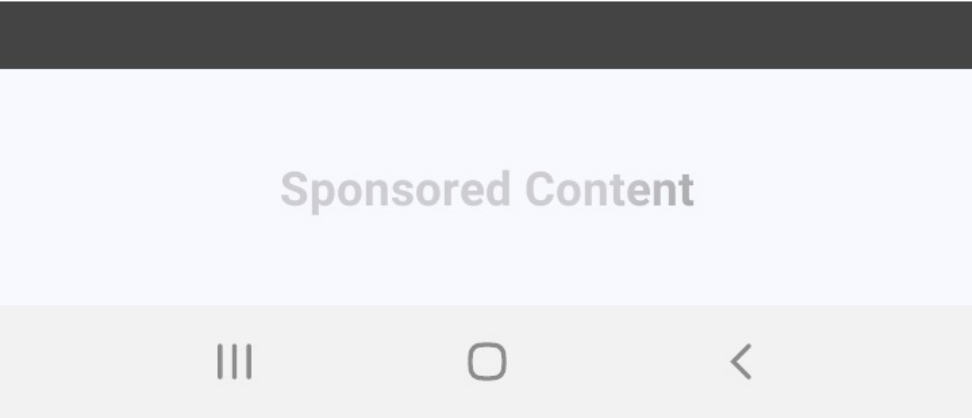
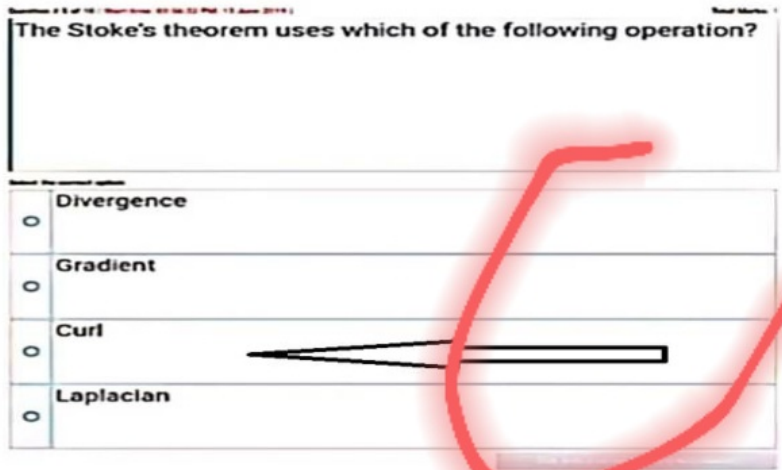
☐	Normal acceleration
☒	Tangential acceleration
☐	Rectangular acceleration
☐	None of these

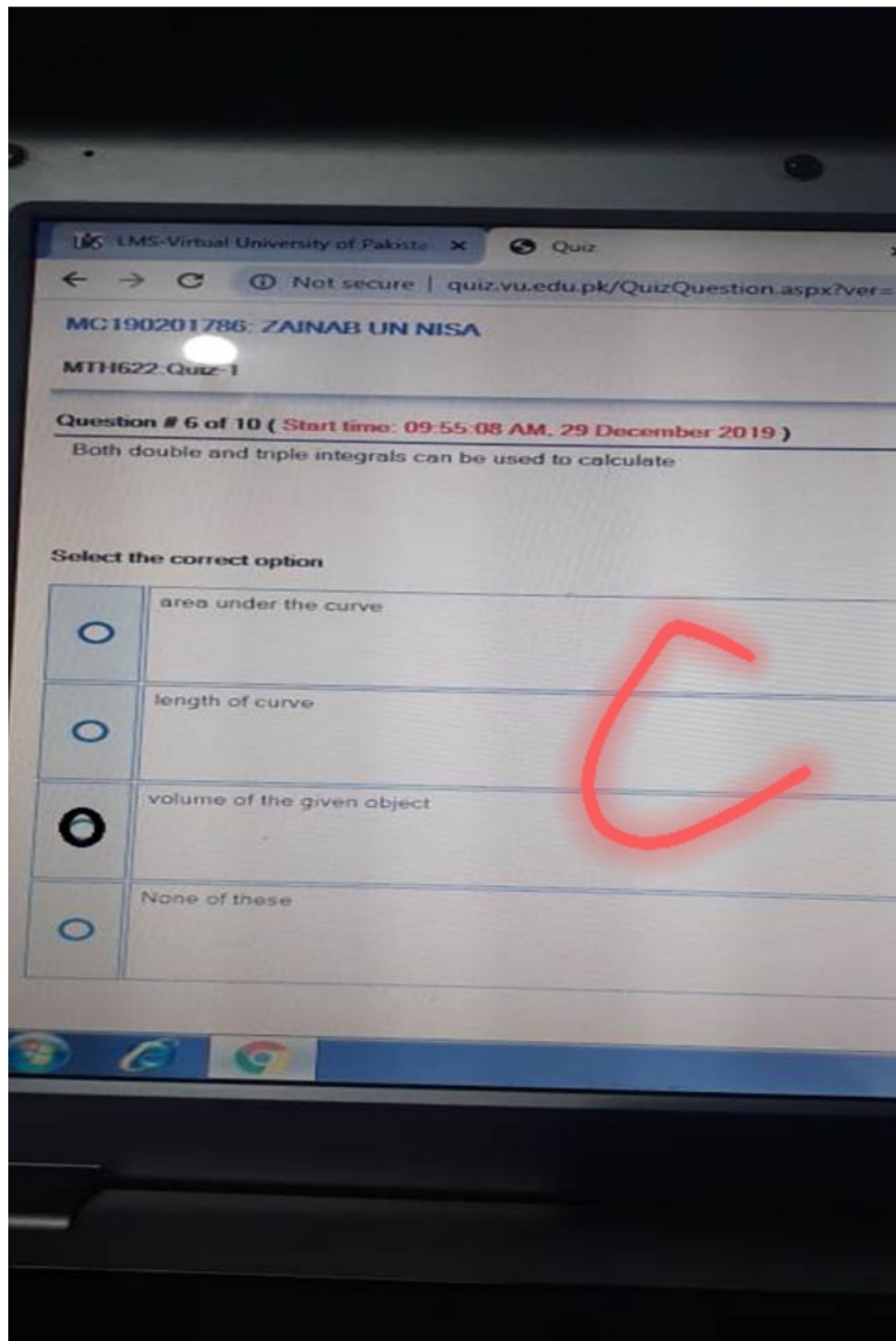
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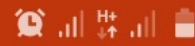






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Quiz
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MTH022:Quiz-1

Quiz Start Time: 10:07 PM, 29 Dec

Question # 1 of 10 (Start time: 10:07:54 PM, 29 December 2019)

To

The radial component of velocity for a particle moving in circular path

Select the correct option



constant



radius itself



variable

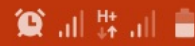


zero

Click to Save Answer & Move to Next Question



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MTH022:Quiz-1

Quiz Start Time: 10:07 PM, 29 Dec

Question # 2 of 10 (Start time: 10:08:50 PM, 29 December 2019)

Tot

What are the rectangular components of velocities in curvilinear motion?

Select the correct option

Reveal Math

☒	$\frac{dx}{dt}$ and $\frac{dy}{dt}$
☐	$\frac{dr}{dt}$ and $\frac{d^2r}{dt^2}$
☐	$\frac{d^2x}{dt^2}$ and $\frac{d^2y}{dt^2}$
☐	None of these

Click to Save Answer & Move to Next Question



MTH622 Vectors and Classical Mechanics

Question No : 20 of 41

Work done is said to be zero, when

Answer (Please select your correct option)

- ☒ a. Some force acts on a body, but displacement is zero
- ☐ b. No force acts on a body but some displacement takes place
- ☐ c. Either a) or b)
- ☐ d. None of these

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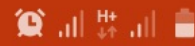
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MTH622: Quiz-1

Quiz Start Time: 10:07 PM, 29 Dec

Question # 4 of 10 (Start time: 10:11:04 PM, 29 December 2019)

To

Tangential component of velocity of the wheel is equal to

Select the correct option

- ☐ angular velocity of the wheel
- ☐ radius of wheel
- ☒ angular velocity of the wheel/radius of wheel
- ☐ angular velocity of the wheel*radius of wheel

Click to Save Answer & Move to Next Question

